–Nabd: An AI-Driven Ecosystem for Maternal Health Monitoring and Risk Detection

***Abstract*—This paper presents Nabd, an AI-driven healthcare ecosystem designed to mitigate pregnancy-related risks and enhance maternal safety. The platform integrates a high-performance .NET 8.0 (ASP.NET Core MVC) back-end with Python-based FastAPI microservices to deliver real-time diagnostic assistance. The system features three primary AI components: an automated OCR pipeline for lab reports and ingredient analysis, a Gestational Diabetes Mellitus (GDM) prediction engine, and a multi-class Maternal Health Risk Detection model. Beyond automated analysis, Nabd provides a comprehensive monitoring framework that facilitates direct communication between doctors and patients, ensuring continuous clinical supervision.**

Experimental results demonstrate exceptional performance across all modules. The Maternal Risk Detection model achieved an overall accuracy of 98%, with high precision and recall across all risk levels. The GDM engine further supports this with a diagnostic accuracy of 93.52% and an ROC AUC of 0.947. Utilizing the docTR framework for OCR and Hugging Face for model hosting, the platform provides a scalable, three-tier architectural solution. By bridging the gap between clinical diagnostics and daily monitoring, the system offers an integrated environment for early risk intervention and maternal health management

Keywords—Maternal Health, Gestational Diabetes, Machine Learning, Risk Detection, Medical Diagnostics, Health Monitoring Systems.

I. INTRODUCTION

Pregnancy is a critical life stage, and focus on maternal and fetal health has grown significantly. Despite this, health risks are rising; a 2021 study [1] found that 14% of pregnant women suffer from gestational diabetes (GDM), which is directly linked to diabetes and cardiovascular diseases. Furthermore, the World Health Organization (WHO) reported in 2023 [2] an alarming increase in maternal mortality and health risks. Beyond medical conditions, concerns regarding skincare and body products are significant. Organizations like the American College of Obstetricians & Gynecologists (ACOG) [3] have warned against ingredients such as retinoids, yet many products lack clear safety information for expectant mothers.

These rising risks stem from unhealthy lifestyles, information gaps, and the lack of reliable alternatives in the absence of direct medical care. Consequently, there is an urgent need for an integrated AI-powered platform that serves as a continuous health monitoring ecosystem. Beyond individual health tracking, a core objective of the system is to assist healthcare practitioners by facilitating seamless, real-time communication between doctors and patients. By providing doctors with an automated analysis of the patient's progress and AI-generated risk alerts, the platform enables more proactive clinical decision-making and ensures that medical supervision is maintained throughout the entire gestational period. Our proposed system, Nabd, addresses these needs by offering AI-based health monitoring, product safety verification, and a direct communication channel for booking and sharing medical reports with physicians.

II. RELATED WORK

The digital healthcare landscape features several specialized commercial platforms, each focusing on a distinct domain:

A. *Clinical Appointment and Communication Systems:* Vezeeta [4] dominates the MENA region as a commercial leader in healthcare scheduling and patient-provider connectivity.

B. *Maternal Health Tracking and Data Logging:*

Flo (Pregnancy Mode) [5] is a global platform providing extensive logging for maternal vitals, symptoms, and fetal development.

C. *Consumer Safety and Ingredient Analysis:*

Yuka [6] serves as a widely adopted tool for evaluating product ingredients via barcode scanning.

While these systems are effective in their domains, Nabd offers a more advanced integration. Unlike Vezeeta's [4] administrative focus, Nabd acts as a clinical decision support system, assisting doctors through continuous monitoring and AI-driven proactive alerts. Furthermore, while Flo [5] acts as a passive tool, Nabd employs an active predictive layer—achieving 98% accuracy in Risk Detection and 93.52% in GDM prediction. Specifically, unlike Yuka [6], which is restricted to a fixed barcode database, Nabd utilizes a flexible OCR-based analysis to interpret any medical report or ingredient label dynamically. This positions Nabd as an intelligent ecosystem bridging the gap between manual tracking and real-time clinical intervention.

III. METHODOLOGY

The proposed system, Nabd, is an integrated web-based ecosystem designed to streamline maternal healthcare by bridging the gap between patients and providers. The system facilitates continuous health tracking and AI-augmented clinical decision-making through four specialized roles: Patients (tracking and AI analysis), Doctors (reviewing results and prescribing), Assistants (scheduling and registration), and

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Administrators (platform oversight). The system's functionality is delivered through a modular architecture, focusing on the following core modules:

A. Medical Value Analysis and Monitoring

This module provides persistent tracking of health parameters through three integrated stages:

1) *Intelligent OCR Integration:*
The system uses the docTR library to read lab reports and Regular Expressions (Regex) to extract specific lab values.

2) *Data Validation and Cleaning:*
To prevent errors, a human verification step is included to confirm the extracted values. Data is then standardized; for instance, inconsistent inputs such as "null," "neg," and "-" are unified into a standard "negative" format before database storage.

3) *Reporting and Rule-Based Alerts:*
Extracted values are used to generate comprehensive reports for physicians. Furthermore, a Rule-Based System (If-Then rules) evaluates if values fall within normal physiological ranges based on clinical standards [7]-[16]. Alerts regarding critical levels, such as glucose or iron, are communicated to the patient in a supportive, non-alarmist manner.

B. Product Ingredient Safety Scanner

To help expectant mothers avoid harmful chemicals in body care products, this module utilizes the docTR library to extract ingredient lists from product packaging. These ingredients are automatically cross-referenced against a medical database compiled from clinical sources [17]-[31], allowing the system to classify each product as 'Avoid' if it contains proven harmful substances, or 'Risky' if the ingredients are medically debated and require professional consultation.

C. AI-Based Maternal Risk Detection

The system prioritizes maternal health monitoring through two specialized Artificial Intelligence (AI) models designed to identify potential health risks during pregnancy.

1) *GDM Prediction Model:*
The model utilizes clinical parameters related to maternal health and applies a Logistic Regression classifier to estimate the probability of developing GDM. The model achieved an overall accuracy of 93.52%, with an ROC AUC score of 0.947, demonstrating strong discriminative capability. This module provides early warnings that can assist physicians in recommending preventive interventions and closer monitoring.

2) *Health Status Classification*
A Random Forest classifier model categorizes maternal health status into three levels: Low Risk, Moderate Risk, and High Risk based on physiological parameters. Experimental evaluation demonstrated an overall accuracy of 98%, with balanced precision and recall across all classes, indicating reliable performance in identifying varying levels of maternal health risk.

D. Clinical Appointments and Personalized Services

To ensure a complete healthcare experience, the system facilitates seamless medical appointment booking with available doctors, while providing a dedicated personalized section where users can manage and store essential information, such as phone numbers and addresses for important places in their pregnancy journey.

E. Physician's Clinical Dashboard

The system empowers doctors with comprehensive management tools by integrating Digital Health Records for reviewing medical histories and trends, an Alert Management system for real-time risk notifications, and Clinic Operations features to oversee patient volume, bookings, and medical team coordination.

IV. SYSTEM ARCHITECTURE AND IMPLEMENTATION.

A. System Architecture

Nabd Platform Architecture

The Nabd platform is built on ASP.NET Core MVC (.NET 8) using a structured three-layer architecture:

- Presentation Layer: Utilizes Razor Views and role-specific layouts for the user interface.
- Business Logic Layer: Encapsulates domain rules within Controllers and dedicated Service classes.
- Data Access Layer: Implements the Repository Pattern to abstract operations for over 20 entities (e.g., IPatient mapped to PatientRepository).

Technical Implementation:

- Dependency Injection: Configured in Program.cs with a scoped lifetime for repositories and services (including AlertService and ProductOcrClient).
- Decoupling: Achieved through constructor injection across all components.
- Infrastructure: Uses a named HttpClient factory for AI services, conventional routing, and standard middleware for HTTPS and static files.

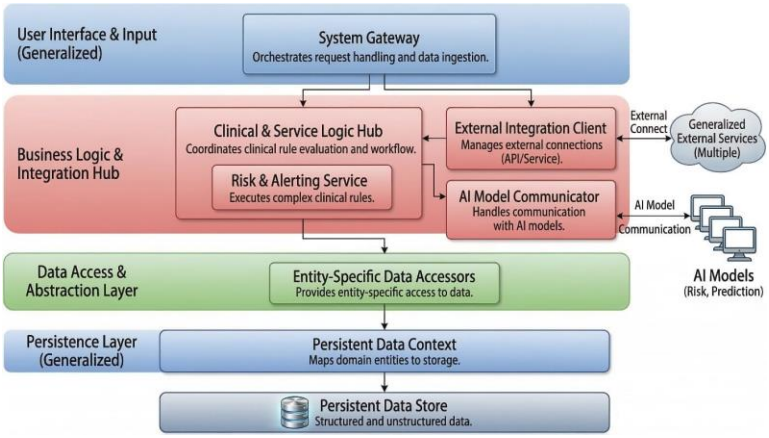

Fig. 1. Nabd System Architecture

B. Technology Stack

The system is built on the following technology stack. The web application framework is ASP.NET Core MVC targeting .NET 8. Data persistence is managed through Entity Framework Core with Microsoft SQL Server as the relational database backend. The Razor view engine renders server-side HTML, augmented with Bootstrap 5 for responsive layout, and vanilla JavaScript with the Fetch API for asynchronous dashboard interactions. Anti-forgery token validation is applied to all state-modifying POST endpoints. Authentication and authorization are implemented using ASP.NET Core Identity, which provides secure password hashing, cookie-based session management, and role claim enforcement. Real-time bidirectional communication for the notification subsystem is delivered via ASP.NET Core SignalR. External AI inference services are implemented as Python FastAPI microservices communicating over HTTP using the IHttpClientFactory-managed HttpClient, with retry logic and exponential back-off.

C. System Modules

The platform is organized into the following major functional modules:

- The Patient Portal, managed by PatientController, provides the patient-facing dashboard displaying pregnancy week, estimated due date, trimester classification, recent blood pressure and blood sugar readings, upcoming appointments, and a chronological activity feed. Patients can log health readings, view prescriptions, manage healthcare location lists, scan product ingredient labels for pregnancy safety, and access a comprehensive medical history timeline aggregating all health events in reverse-chronological order.

- The Doctor Dashboard provides doctors with a consolidated view of their assigned patients, including AI-analyzed laboratory results, ultrasound reports, prescriptions, and clinical notes.

- The Assistant Dashboard, served by AssistantController, delivers a clinic-level operational view including appointment schedules and workload statistics per doctor, with heavy data loaded asynchronously via AJAX.

- The Admin Portal provides system administrators with platform-wide oversight, including user account management, clinic registration, doctor verification, and system monitoring. The Appointment and Booking Management module enables conflict-free real-time slot reservation enforced at the database level through a unique index on AppointmentID.

D. Authentication and Authorization

The system implements authentication and authorization using ASP.NET Core Identity integrated with the existing custom User and Role data model. The User entity stores a hashed password in the PasswordHash field, managed by Identity's PasswordHasher, along with an email address used as the unique login identifier. Role-Based Access Control (RBAC) is enforced using Identity's built-in claims and policy middleware, mapping the existing role definitions — Patient, Doctor, Assistant, and Admin — to route-level and action-level access restrictions via the [Authorize(Roles = "...")] attribute. Upon successful login, ASP.NET Core Identity issues an encrypted authentication cookie bound to the user's role claims, which is validated on each subsequent request by the authentication middleware configured in Program.cs. Patients are restricted exclusively to patient-facing routes, doctors to clinical management routes, assistants to clinic operation routes, and administrators to the admin portal, preventing any cross-role data exposure. CSRF protection is uniformly applied via [ValidateAntiForgeryToken] on all POST endpoints throughout the application.

E. AI Integration within the System

AI capabilities are integrated into the platform through RESTful API communication with externally hosted Python FastAPI microservices. For laboratory test analysis, scanned images of printed test reports are submitted to an OCR pipeline employing the DocTR deep learning framework for text detection and recognition, followed by regular-expression-based parameter extraction for eight supported test types: CBC, Urinalysis, HbA1c, TSH, Ferritin, Blood Group, HBsAg, and HCV. Extracted parameter values and clinical status classifications are returned as structured JSON and persisted in the LabTest.AI_AnalysisJSON database field. For product safety scanning, the ProductScanController dispatches ingredient label images to a Hugging Face-hosted OCR service via ProductOcrClient, which classifies each detected ingredient as safe, risky, or to be avoided during pregnancy. All AI communication channels use the IHttpClientFactory pattern with configurable timeouts and automatic retry behavior.

F. Notification and Alert System

The system employs a dual-layer notification mechanism. The first layer is a server-side threshold evaluation engine implemented in the AlertService class, which persists structured Alert records to the database whenever patient health data crosses clinically defined boundaries. Blood pressure readings trigger danger alerts for critically elevated values and warning alerts for values indicative of hypertension. Blood sugar readings generate alerts for hypoglycemia, elevated fasting glucose, and critical hyperglycemia. Additional alerts are generated from abnormal laboratory AI JSON analysis results, lifestyle risk flags (smoking, alcohol use), and upcoming appointment reminders within a 24-hour window. A daily deduplication mechanism prevents redundant alert records. The second layer is a real-time push notification system built on ASP.NET Core SignalR, which delivers instant in-browser notifications to connected users when new alerts are generated, appointment statuses change, or critical health thresholds are crossed, without requiring a page refresh. The SignalR hub broadcasts role-scoped notifications,

ensuring that alert events are dispatched only to the relevant connected client.

G. User Interface Design

The user interface is implemented using Bootstrap 5 in conjunction with custom CSS scoped to each role-specific Razor layout file: _PatientLayout.cshtml, _DoctorLayout.cshtml, _AssistantLayout.cshtml, and _AdminLayout.cshtml. These layouts define role-specific navigation bars, sidebar structures, and notification panel components, ensuring visual consistency within each user context. Patient-facing views include interactive health tracking panels, pregnancy progress indicators, a chronological medical history timeline, appointment calendar components, and the product safety scanner. The assistant dashboard uses asynchronous AJAX requests to populate schedule grids and summary statistics, reducing time-to-first-meaningful-paint. The admin portal presents tabular management interfaces for users, clinics, and system metrics. All layouts are fully responsive, adapting to varying viewport sizes using Bootstrap's grid system. View models such as PatientDashboardViewModel, AssistantDashboardViewModel, and MedicalHistoryViewModel decouple data preparation from view rendering, following established separation-of-concerns principles.

H. Security and Data Management

The system enforces multiple layers of security to protect sensitive medical data. Password security is managed by ASP.NET Core Identity's PasswordHasher, which uses PBKDF2 with HMAC-SHA256 and a randomly generated salt per user, ensuring resistance to both brute-force and rainbow-table attacks. Input validation is enforced through ASP.NET Core model binding, data annotation attributes, and explicit server-side checks within controller actions. CSRF protection is uniformly applied via [ValidateAntiForgeryToken] on all POST endpoints. HTTPS redirection is enforced for all non-development environments, and HTTP Strict Transport Security (HSTS) is enabled to prevent protocol downgrade attacks. Role-based data access is enforced at the controller layer through Identity's authorization policies, ensuring that patient, doctor, assistant, and admin identifiers are validated before any query executes, thereby preventing unauthorized cross-user data access. At the database level, Entity Framework Core enforces referential integrity through explicitly configured cascade behaviors, unique constraints, and restricted delete operations. Medical image file uploads enforce content-type allowlisting and a maximum size limit before any external service invocation, preventing malformed or oversized payloads from reaching the AI pipeline.

V. MATHEMATICAL MODEL

The analytical core of Nabd is governed by distinct mathematical frameworks tailored for predictive accuracy and logical classification.

A. GDM Probability Model

The risk assessment for GDM is modeled using a logistic regression framework to determine the probability P based on clinical features:

$$P = 1 / (1 + \exp(-z)) \quad (1)$$

where z represents the weighted sum of input variables:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n \quad (2)$$

In (2), β_0 is the intercept, $\beta_1, \ldots, \beta_n$ are the regression coefficients, and $x_1, \ldots, x_n$ are the patient's medical parameters.

B. Maternal Health Risk Classification

To categorize overall health status into Low, Moderate, or High risk, a Random Forest ensemble is utilized. The final classification $\hat{C}$ is determined by aggregating the predictions of N individual decision trees $\{T_1, T_2, \ldots, T_n\}$ through majority voting:

$$\hat{C} = \text{mode}\{T_1(x), T_2(x), \ldots, T_n(x)\}. \quad (3)$$

For each tree, the optimal split at each node is determined by minimizing the Gini Impurity G:

$$G = 1 - \Sigma\, p_j^2, \quad (4)$$

where p_j is the probability of a sample belonging to class j at a specific node.

C. Product Ingredient Safety Mapping

The ingredient safety classification is defined by a mapping function $f(I)$ that evaluates the set of extracted ingredients I against specialized medical datasets D. The classification logic is represented as:

$$\begin{aligned} f(I) &= \text{Avoid, if } (I \cap D_harmful \neq \emptyset), \quad (5) \\ f(I) &= \text{Risky, if } (I \cap D_debated \neq \emptyset), \\ f(I) &= \text{Safe, otherwise.} \end{aligned}$$

VI. EXPERIMENTAL SETUP

A. Data Profile:

Maternal Risk Dataset: 30,041 clinical records categorized into High, Moderate, and Low risk levels, and GDM Dataset: 3,098 samples focusing on metabolic indicators and glucose parameters.

B. Data Preprocessing & Balancing:

- Normalization: Numerical features were scaled using StandardScaler, with OneHotEncoding applied to categorical variables.
- Imbalance Handling: Various resampling and data-balancing techniques were evaluated to address minority class representation. The optimal method was selected based on achieving the highest balance between Sensitivity and Precision.
- Validation: A stratified 80/20 split was maintained for training and testing.

D. Specialized Model Configurations

- Predictive Engines: Benchmarked Logistic Regression and Random Forest (200 estimators). A specialized probability threshold of 0.7 was implemented for GDM prediction to ensure clinical reliability and minimize false positives.

- OCR Pipeline: Integrated the docTR framework for high-accuracy text extraction, with custom processing logic to map outputs to standardized clinical databases.

C. Hardware & Software Environment

-Nabd Framework: The platform's ecosystem was developed using .NET 8.0 (ASP.NET Core MVC), ensuring a high-performance environment for clinical data management and user role orchestration.

-AI Compute Power: Model training and inference were executed on Google Colab, leveraging TPU v4 accelerators to optimize processing speeds for large-scale datasets.

-Development Stack: The predictive engines were built using Python 3.10, with Scikit-learn for machine learning pipelines and Pandas for efficient data manipulation.

-System Integration: A FastAPI layer was utilized to bridge the .NET back-end with the Python-based AI models, enabling real-time, asynchronous communication through RESTful APIs.

VII. RESULTS AND DISCUSSION

The evaluation of Nabd demonstrates the clinical efficacy and technical stability of the integrated ecosystem.

A. Predictive Engine Performance

The system's AI layer showed exceptional diagnostic reliability across both predictive engines:

- Maternal Health Risk Detection: The Random Forest model achieved an outstanding accuracy of 98%, effectively categorizing patients into High, Moderate, and Low risk levels. The high precision across all classes ensures that the Physician's Dashboard receives reliable, actionable alerts.
- GDM Prediction: The Logistic Regression engine achieved an accuracy of 93.52% and a ROC AUC of 0.947. With a Recall of 77.7%, the system prioritizes identifying potential cases for early intervention, a critical requirement in proactive screening.

B. Automated OCR & Real-time Processing

The docTR pipeline transformed unstructured medical data into standardized clinical records with high recognition accuracy. The integration with FastAPI enabled low-latency ingredient analysis, where the system successfully cross-referenced extracted text against the safety database to provide immediate "Avoid/Safe" classifications.

C. System Discussion

The results validate the. NET-Python hybrid architecture. While the GDM model's precision (0.70) reflects a trade-off for higher sensitivity, this is a deliberate system design to ensure no high-risk case is overlooked. The seamless data flow—from image upload via the web interface to risk assessment and SignalR alert triggering—demonstrates the platform's readiness for real-world clinical deployment.

VIII. CONCLUSION

This research presented Nabd, a comprehensive, multi-layered healthcare ecosystem designed to bridge the gap between passive tracking and active clinical intervention. Built on .NET 8.0 and ASP.NET Core MVC, the platform leverages a robust three-tier architecture and the Repository Pattern to ensure scalability and maintainable decoupling. Security and real-time responsiveness are central to the design, utilizing ASP.NET Core Identity for secure role management and SignalR for instantaneous risk notifications.

The integration of FastAPI microservices empowers the platform with advanced diagnostic capabilities. By achieving a 98% accuracy in Maternal Risk Detection and 93.52% in GDM prediction, the system provides a reliable decision-support tool for healthcare practitioners. Coupled with the docTR-based OCR pipeline, Nabd automates the labor-intensive task of medical report analysis and ingredient safety verification. Ultimately, this research demonstrates that the convergence of modern software engineering and deep learning creates a scalable, high-integrity environment for mitigating pregnancy risks and improving maternal health outcomes.

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